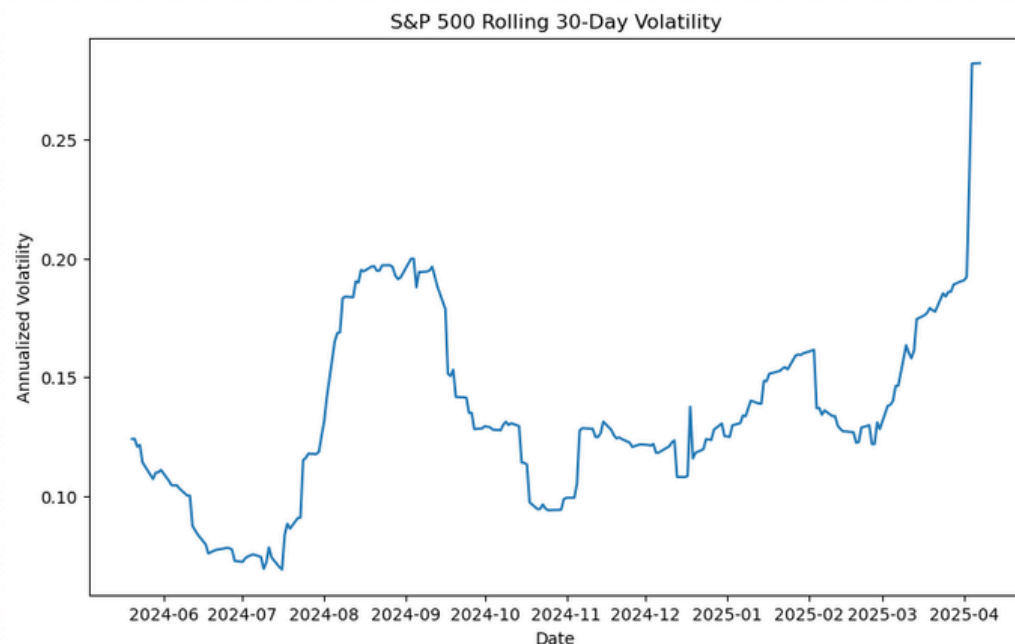


## Talk vs. Trends: Sentiment & Alignment Analysis of News Providers Tweets vs. Real Market/Macroeconomic Outcomes.

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### Background



**Figure 1. Motivation.** In today's interconnected markets, news and events spread instantly on social media, affecting sentiment and asset prices, with market volatility reaching record high-levels. By analyzing influential tweets from sources like the Wall Street Journal (WSJ) and Bloomberg and correlating sentiments with market indicators (e.g., the S&P 500), investors can track how sentiment shifts during volatile periods. Our integrated dashboard combines tweet data, sentiment analysis, and market metrics, enabling users to spot potential patterns, compare sentiment trends to market performance, and make more informed investment decisions.

### Contributions Summary

**Arnav Mehta:** Conducted preliminary research on potential APIs to use. Collected and scraped tweets to create 10 csv files and assisted in creating the NLP analysis

**Nico Poulsen:** Assisted in the collection and scraping of tweets to create the 10 csv files and created the NLP analysis algorithm.

**Mark Abousleiman:** Created the sankey visualization of news sources and common words used.

**Talal Fakhoury:** Created the 2<sup>nd</sup> visualization on the average sentiment scores per news source tweets, and 3<sup>rd</sup> visualization of subplots.

### Process & Method

#### Data Sourcing and Collection

Collect the last 200 tweets from major news providers and gather market data (e.g., S&P 500) leveraging twitterapi.io

#### Data Cleaning

Remove noise (punctuation, hashtags) from tweet texts and apply stopwords filters.

#### Visualization and Analysis

Create interactive plots (Sankey diagrams, time series overlays) to explore sentiment vs. market trends.

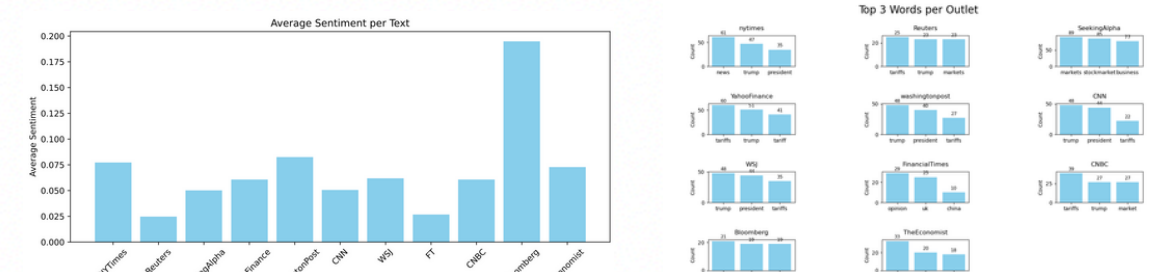
#### Sentiment Analysis

Use NLP methods NLTK for stopwords and TextBlob for its built in sentiment analyzer to compute sentiment scores for each grouped topic

### Key Findings

### Link to our sankey diagram!

```
== CNN ==
inflation: 0.359
interest rates: 0.183
recession: -0.191
sp500: None
jobs: 0.083
unemployment: 0.25
economy: 0.049
growth: None
federal reserve: 0.183
monetary policy: None
tariffs: 0.065
trade: 0.02
gdp: None
housing: None
trump: 0.028
deficit: None
debt ceiling: None
consumer spending: None
yields: None
banking: None
```



Within the visualizations, we see frequent references to key terms like “inflation” “tariffs” and “trump” highlighting how fast-evolving media coverage focuses on economic uncertainty. Furthermore, we see the differences of opinions and coverage across major news providers. For example CNN is relatively more positive in terms of inflation in comparison to other providers. On the other hand, they have negative sentiment about recession with -0.191 which aligns with the recent worries on tariffs and its long term impact on the economy.

The enhanced insights gained from these news-driven sentiments can support more informed investment decisions by highlighting emerging risks and market opportunities. Additionally, we can integrate machine learning techniques to forecast sentiment shifts and identify patterns over time, paving the way for advanced predictive analytics in financial markets.